

Project Synopsis
on
Social Media Sentiment Analysis

Submitted as a part of course curriculum for

Bachelor of Technology
in
Computer Science & Engineering



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DECLARATION

We hereby declare that this submission is our work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgement has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled “Social Media Sentiment Analysis” which is submitted by **Siddharth Kesharwani, Shubham Choudhary, Yash Gupta, Vanshita** in partial fulfilment of the requirement for the award of degree

B. Tech. in the Department of Computer Science & Engineering of Dr A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Date:

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ABSTRACT

Some of the most amazing discoveries in this decade have been made thanks to technology and technological advancements. The world seems to get better every day thanks to the rapidly evolving technology, systems to support them, and back-end processing power.

Social media sentiment analysis is currently one of the most challenging problems. Sentiment Analysis is the process of estimating the emotional tone and opinion behind online text across platforms and demographics. Rapid opinion shifts, sarcasm, and platform-specific language are just a few examples of challenges. The main objective of the analysis process, which enhances brand engagement and strategy development, is to deliver an accurate and detailed understanding of public perception in real-time.

Sentiments about various aspects like product quality, service, or pricing can be captured and studied using this technique. Brand managers can monitor emerging trends and respond quickly using real-time tracking. Influencer activity and its impact on brand image can also be analyzed. Competitor sentiment comparison helps companies refine strategies. To accomplish the mentioned goals, we are using Natural Language Processing(NLP) models like BERT and RoBERTa. Our use of dashboard technologies such as Dash and Streamlit embedded with PostgreSQL databases enables us to produce smooth and insightful visualizations for drawing critical conclusions. The biggest challenge facing the sentiment analysis industry today is obtaining clean, unbiased, and relevant data in real-time. Because in this scenario, every sentence, phrase, and even emoji contributes crucial information to the model's performance. Therefore, social media sentiment analysis is a difficult task to complete, which will also help solve many real-world and industrial problems. For example, because of abrupt sentiment shifts, it is difficult to detect early signs of brand crises without advanced monitoring. In a similar vein, a model in the marketing field allows quick and accurate visualization of customer feedback trends. The training of numerous economic projects, including brand reputation management and influencer marketing analysis, will undoubtedly be helpful to the industry.

CHAPTER 1: Introduction

1.1 INTRODUCTION:

Technology and Technological developments in this decade have led to some of the most awe-inspiring discoveries. With rapidly changing technology and systems to support them and provide back-end processing power, the world seems to be becoming a better place to live day by day.

One of the biggest challenges of recent times is Social Media Sentiment Analysis. Sentiment Analysis is **the operation of capturing emotions, opinions, and attitudes from online text across different platforms and demographics**. Challenges may come in many forms, such as rapid opinion shifts, sarcasm, platform-specific language, and data noise. The sentiment analysis process improves brand understanding, and the main goal is to restore the true public perception as it evolves in real-time.

Sentiment analysis plays an important role in the real-life marketing field, where brand sentiments related to product quality, service, or pricing can be captured and studied through the process. Real-time tracking of public opinion can help in crisis management, influencer impact can be analyzed for brand strategy, and competitor comparison can guide companies to refine their strategies.

Here, we are using the Natural Language Processing (NLP) models like BERT and RoBERTa to achieve the mentioned goals. Our use of dashboard technologies such as Dash and Streamlit embedded with PostgreSQL databases helps us to generate a fine-tuned and smooth visualization of live data, which can be further used to draw critical insights.

1.2 PROBLEM STATEMENT:

To achieve a high quality sentiment interpretation without bias, sarcasm misinterpretation, and platform-specific noise is the biggest challenge in today's Natural Language Processing industry. Because the training of a complete model in such cases depends on every sentence, phrase, and even emoji from the text data. Hence, social media sentiment analysis is the biggest task to achieve which further helps to solve various real-world and industrial problems such as in real world — abrupt shifts in public sentiment make it difficult for brands to detect crises early, similarly in the marketing field the customer feedback trends can be visualized easily and effectively by the model. It will certainly help industries to train various economical projects like – brand reputation management, influencer marketing analysis, customer engagement prediction, and many more.

1.3 OBJECTIVE:

Objective is to build a website using Streamlit where the process of social media sentiment analysis can be carried out, and insights from different platforms can be generated that provide a better view to the user. A real-time social media dataset will be uploaded by the user, and in return, it will provide detailed sentiment analysis with aspect-based categorization.

The model is efficient enough to perform multi-platform sentiment tracking, aspect-based analysis, and comparative brand monitoring efficiently.

1.4 SCOPE:

Nowadays industries are crazy about real-time sentiment analysis tools which can be utilized by them to track brand perception for different purposes like customer engagement, crisis detection, marketing strategy planning, and many more, but the limit is clean and structured data here. NLP models like BERT along with real-time streaming APIs can be used to collect, classify, and analyze sentiments efficiently. The generated insights are realistic and this processed data can be used in many ways like local brands, which are unable to afford expensive market research, can rely on sentiment tracking to design their campaigns. This method provides a faster and cost-effective solution without dependency on large survey agencies. Many other industrial applications also sustain for the same.

CHAPTER 2: Literature Review

Cardiffnlp et al. [1] have developed a RoBERTa-base model specifically trained on approximately 58 millions tweets and fine-tuned for sentiment analysis using the TweetEval benchmark. The authors categorized text into three sentiment classes: negative, neutral, and positive. According to the study, the model demonstrated the effectiveness of transformer-based models for social media text analysis, particularly for platforms like Twitter. The researchers highlighted that the model can analyze brand mentions and consumer opinions, forming a foundation for multi-platform sentiment analysis projects. The study concludes by mentioning the continuous improvement of their models with updated datasets.

Repustate [2] has explored real-time sentiment analysis as a machine learning technique to automatically detect and extract sentiment from text data as it occurs. Their research emphasized the importance of multi-platform capability, including text, audio, and video formats, for a complete social media monitoring solution. The authors demonstrated that real-time tracking helps brands detect threats, address negative comments swiftly, and manage brand mentions effectively across multiple platforms like Twitter, Instagram, and Facebook.

Sonarplatform [3] introduced aspect-based sentiment analysis as a refinement over traditional techniques. The researchers explained that it allows deeper insight by distinguishing sentiments attributed to specific aspects like product quality, service, or pricing. An example provided was a restaurant review showing different sentiments towards service and food quality. The study concluded that aspect-level analysis enables businesses to derive actionable insights from complex reviews, providing a strategic edge in customer satisfaction analysis.

EELET [4] conducted a study on the influence of social media influencers on brand loyalty. The finding revealed a mean response score of 4.20690 on a 5-point scale, showing a significant impact ($t= 48.915$, $p < .05$). The authors observed that platforms like Instagram and YouTube had the highest usage for influencer content. This research demonstrated that influencer marketing significantly enhances brand perception and loyalty, offering brands a potent strategy for sentiment building.

SSRN [5] presented a fusion model architecture combining BERT and RoBERTa for enhanced analysis of tweets, particularly during the COVID-19 pandemic. The researchers reported achieving high F1-scores, recall, and accuracy across sentiment classes. The study highlighted the importance of preprocessing and noise elimination to maintain model accuracy. They concluded that fusion architectures improve classification robustness, offering better results for brand monitoring on social media.

Sprinklr [6] discussed the features necessary for effective competitor analysis in social media sentiment tracking. Their research emphasized the importance of broad data coverage across news, blogs, forums, and social channels. The study noted that AI-driven insights allow brands to forecast trends, react quickly to competitor moves, and maintain a competitive advantage.

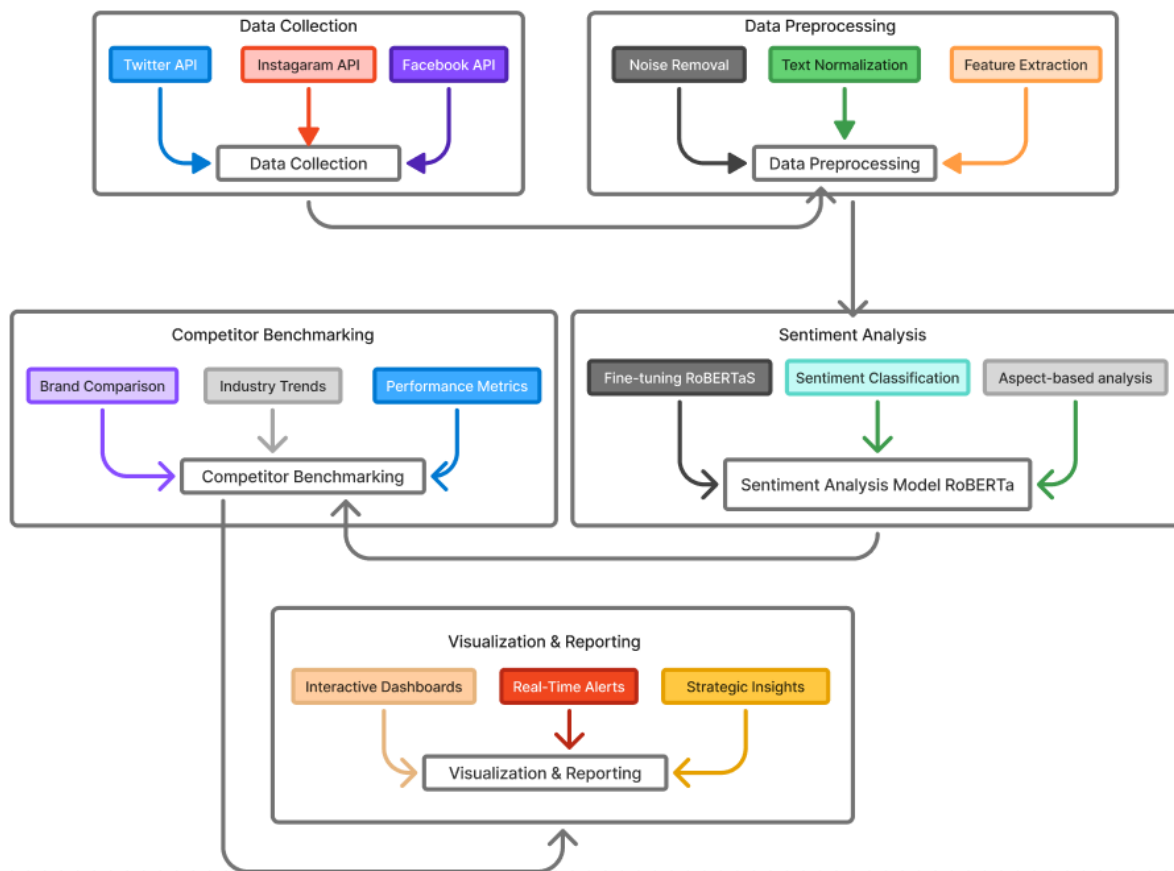
Convin [7] evaluated sentiment analysis platforms like Brandwatch and Talkwalker. According to their research, AI-driven sentiment tools provide comprehensive insights into audience emotions across multimedia content, including images and videos. The study underlined the necessity of multimedia sentiment tracking and benchmarking features to monitor competitor performance effectively, enabling brands to make informed strategic decisions.

CHAPTER 3: Proposed Methodology

3.1 Flowchart:

Our complete study and deployment are based on transformer-bases models fine-tuned for sentiment analysis. RoBERTa and BERT-bases architectures are used to classify social media posts into positive, negative, and neutral sentiments.

First, data is collected from multiple platforms like Twitter, Instagram, and Facebook. The collected data is preprocessed to remove noise and irrelevant content. After preprocessing, the sentiment analysis model (RoBERTa) is trained and fine-tuned. The fine-tuned model is then used to perform competitor benchmarking by analyzing sentiments related to different brands. Insights generated from this analysis are visualized through dashboards to help brands make strategic decisions based on public sentiment.



3.2 Algorithm Proposed:

Dataset: For this study and training purpose, a custom dataset was created comprising approximately 10,000 social media posts collected from platforms such as Twitter, Instagram, and Facebook. Each post includes the text content, timestamp, platform source, and associated brand or competitor tag. The data was preprocessed to remove irrelevant entries, clean noise (hashtags, mentions, special characters), and normalize the text to ensure consistency and enhance model performance.

Sentiment Analysis using RoBERTa: Sentiment analysis of user-generated content is crucial for competitor benchmarking and brand reputation monitoring. In this study, the RoBERTa (Robustly optimized BERT approach) transformer model is employed to classify sentiments into three categories: positive, negative, and neutral. The major steps involved in building the sentiment analysis system are:

- Initially, the collected text data is tokenized and encoded using RoBERTa's tokenizer, preserving the context and meaning.
- A pre-trained RoBERTa model is fine-tuned on our labeled dataset to adapt it specifically to social media sentiment classification.
- Hyperparameters such as learning rate, batch size, and number of epochs were optimized to achieve the best performance.
- The fine-tuned model is evaluated on a validation set to ensure generalization and reduce overfitting.

Competitor Benchmarking: Once the sentiment analysis model is fine-tuned and evaluated, it is further used for competitor benchmarking. This involves analyzing the sentiment associated with each brand's mentions, and the results are aggregated to form comparative insights.

Visualization and Deployment: To make the outputs easily interpretable, interactive dashboards and visual analytics are generated using tools like Streamlit and Plotly. The trained model is deployed through a Streamlit-based user interface where users can input text data, view sentiment predictions, and visualize competitor sentiment comparisons dynamically.

CHAPTER 4: Technology Used

Technologies used for the implementation of the model are:

- 1) Natural Language Processing (NLP)
- 2) Deep Learning
- 3) Transformer Models (RoBERTa)
- 4) Sentiment Analysis
- 5) Data Visualization (Matplotlib)

Needed:

- 1) Google Colaboratory
- 2) Visual Studio Code (VS Code)

Requirements:

- 1) GPU for fine-tuning RoBERTa
- 2) Internet access for data collection and model deployment

CHAPTER 5: Conclusion

Currently, the social media landscape plays a crucial role in shaping brand perception. Platforms like Twitter and Instagram generate massive amounts of user-generated content that can be utilized for brand monitoring, competitor benchmarking, and strategy development. However, the sheer volume and unstructured nature of social media data pose significant challenges.

This project addresses these challenges by building a Social Media Sentiment Analysis system capable of real-time tracking, multi-platform analysis, and influencer impact measurement. By leveraging NLP models such as RoBERTa for fine-tuned sentiment classification and utilizing tools like Streamlit and PostgreSQL for dashboard visualization and data storage, the system delivers technical depth and actionable insights.

Aspect-based sentiment analysis further refines the understanding of user opinions across various parameters such as product quality, service, and pricing. Additionally, competitor comparison features enable brands to benchmark their performance relative to rivals over time, offering strategic advantages.

Thus, the project presents a scalable, insightful, and practical solution for brands seeking to enhance their online presence, monitor customer sentiment effectively, and shape their marketing strategies accordingly.

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